Chapter 7, Problem 1RQ

Problem

An *RC* circuit has $R = 2\ \Omega$ and $C = 4$ F. The time constant is:

(a) 0.5 s

(b) 2 s

(c) 4 s

(d) 8 s

(e) 15 s

Step-by-step solution

Step 1 of 1

Calculate the time constant for an *RC* circuit using the following formula.

$$\tau = RC$$

Substitute $2\ \Omega$ for resistance *R* and $4\ \text{F}$ for capacitance *C*.

$$\begin{aligned}\tau &= 4(2)\\ &= 8\ \text{s}\end{aligned}$$

Thus, the value of the time constant for an *RC* circuit is $8\ \text{s}$. Hence the correct option is $\boxed{\text{d}}$.

As the option d is correct remaining options are not suitable for the correct option.